

ENERGY EFFICIENCY IN HISTORIC HOMES

There are a few hurdles to overcome in retrofitting historic homes, such as staying out of the way of visitors, but the effort can lead to greater energy efficiency, greater comfort, and better protection for precious artifacts.

**BY JAMES CAVALLO
AND CHET MARSZALEK**

Reducing the amount of energy used in historic homes has two big benefits. First, it can help government agencies or private conservators to stretch tight budgets. Second, and almost as important, the efficiency improvements may extend the lives of important historic properties and can help to maintain a better indoor environment for the artifacts housed within them.

The Road to Galena

Visitors to a historic home probably don't think first about saving energy. Yet that was the first thought on the mind of a team who visited three historic homes in Galena, Illinois, in the early summer of 2003. The Illinois state energy office, which is housed within the Department of Commerce and Economic Opportunity, working with the Illinois Historic Preservation Agency, wanted to learn whether the tools and methods of home energy ratings were applicable to historic buildings. Two years earlier, these tools and methods had been successfully applied to older homes in Quincy, a historic Illinois community along the Mississippi River. Now John Marley of the Illinois energy office wanted to see what energy rating could offer in state-owned historic homes.

In the fall, winter, and spring of 2001–02, several energy raters working for Adams Electric Cooperative had conducted home energy ratings to assist



The home of President U.S. Grant in Galena, Illinois, is visited by 65,000–85,000 people each year.

homeowners in and around Quincy. The approach taken was to do HERS ratings for the homes, then work with the homeowners to define possible energy improvements for the property, and rate the homes again to estimate potential energy savings.

The state energy office provided an incentive for the homeowners to improve their properties by offering them \$50 per rating point improvement. Thus, if a home scored a 55 in its initial configuration and could achieve a 20-point improvement in its HERS score by installing attic and wall insulation and a new furnace, the homeowner would receive \$1,000 after those improvements were completed and verified. Generally, the incentives covered 25%–50% of the improvement costs, but the energy savings of the improvements covered the remainder within three to

four years. The energy raters found that the initial ratings averaged 55. The average rating point improvement from the installed energy efficiency measures was 22 (see “How the Good Old Days Weren’t,” *HE* Sept/Oct '02, p. 16).

The Quincy project was completed as the Illinois state budget was drifting into serious deficit. Interest at the energy office turned to possible applications of the known cost-cutting techniques in state-owned buildings. The lessons of Quincy seemed applicable to the historic state-owned homes in Galena.

Galena is a town rich in historic buildings. Like Quincy, Galena grew up in the mid-nineteenth century. The source of its early wealth was its lead mines. From the late 1830s until the 1890s, the town prospered. But Galena's fortunes reversed quickly at the close of the century, and for much of the twenti-

eth century the town lay dormant, its homes and shops virtually frozen in time, until tourism brought a rebirth in the late 1960s and 1970s.

One of the principal points of interest for visitors to Galena is the home of President U.S. Grant. Grant moved to Galena in 1860 to work with his brothers in the family's leather goods store, having retired from the army several years before. With the start of the Civil War, Grant returned to military life, and achieved his legendary victories. Following the war, a group of local businessmen and Congressman Elihu Washburne purchased an Italianate-style brick home and presented it to General Grant. The home remained in the Grant family until 1904, when it was given to the city of Galena. Ownership was transferred to the state of Illinois in 1931, when Galena suffered financial difficulties.

To organize the work that the Illinois energy office envisioned at the historic homes in Galena, John Marley contacted Mike Jackson, chief architect at the Illinois Historic Preservation Agency (IHPA) and Terry Miller, manager for the IHPA Galena Historic Site. Three houses were chosen for examination. These were the Grant home, the home of Congressman Washburne (where Grant waited one evening to hear the results of the election of 1868); and the Callahan house (a house built in 1891 that is currently used as IHPA offices for the Galena Historic Site). Fortunately, all three homes had been retrofitted with modern forced-air furnaces and central air conditioning.

Searching for Efficiency Opportunities

A team of raters, including ourselves, began by performing home energy ratings on the three historic homes. Our goal was to identify cost-effective energy efficiency measures that could be implemented within the restrictions imposed by historic preservation. We could not alter the exterior appearance of the structures, and all building materials had to be true to the period. Nor could we

	Conditioned Space (ft ²)	Baseload (therms/day)	Space Heating (Btu/HDD/ft ²)
Grant Home	3,170	0.56	16.3
Washburne House	4,290	0.00	14.2
Callahan House	1,520	0.00	8.1

Note: Table 1 is derived from analyzing the natural gas bills.

	Conditioned Space (ft ²)	Blower Door Results (CFM ₅₀)	Estimated Space Heating Usage (therms/yr)	Estimated Space Heating Cost (\$/yr)	HERS Score
Grant Home	3,170	4,570	2,860	1,080	70
Washburne House	4,290	7,260	4,820	1,700	61
Callahan House	1,520	2,370	1,130	460	77

Note: Table 2 is from the REM/Rate output taking the houses as we originally found them.

alter the interior spaces open for public viewing. Even with all these restrictions, the homes still offered important opportunities for efficiency improvement.

In looking for cost-effective opportunities, we began by examining past utility bills for the three homes. Since Galena's harshest weather comes in the winter, with its design temperature of -11°F and its 7,376 expected annual heating degree-days, we focused on the natural gas bills. These bills were analyzed to break out an estimate of the energy used for space heating and for baseload consumption (see Table 1). ("Screening Before Rating," p. 36, describes an easy-to-use method for separating natural gas utility bills into baseload and space-heating load.)

The utility bills represented in Table 1 include the period of July 2002 through June 2003. The table shows no baseload gas use for the Washburne and Callahan houses. This is because there were no gas appliances in these homes other than the forced-air furnace. The Grant home had a water heater and a small kitchen that also served as a staff

room. Based on previous experience, the energy analysts expected that baseload use would fall in the range of 0.5 to 1 therm per day. Estimates above 1 therm per day suggest a need to investigate the possible presence of an inefficient water heater, a leaking faucet, or a poorly operating gas appliance. The Grant home measured well below this level of baseload gas use.

As a rule of thumb, homes with space-heating use above 10 Btu per heating degree-day per ft² (Btu/HDD/ft²) of conditioned space tend to provide fertile ground for cost-effective energy savings. An analysis of utility bills cannot indicate the best retrofits for energy savings, such as air sealing, adding insulation, upgrading mechanical systems, or tightening leaky ducts that pass through unconditioned areas. But the high space-heating consumption of two of the historic homes, adjusted for the size of the home and the weather conditions, pointed to the potential for efficiency improvements.

To investigate specific energy improvements, each of the three homes



ILLINOIS HISTORIC PRESERVATION AGENCY

The home of Congressman Elihu Washburne is where President Grant played cards as he waited for election returns to be telegraphed in November 1868.

was given a home energy rating, using HERS. Measurements and calculations for the rating included the floor area and volume of the conditioned space, exterior wall dimensions, ceiling area, insulation levels and types of insulation, and the sizes of the windows and exterior doors. We also gathered nameplate information on the efficiencies of the space-heating, air conditioning, and water-heating systems. The collected information was later entered into REM/Rate, an energy rating software tool (see Table 2).

The Grant and Callahan houses had energy rating scores that were higher than the rating scores of more than 80% of Quincy homes. The Washburne home was also above the average rating for homes in Quincy. The analysts found that the source of the higher performance was the heating systems that had been installed in the Galena houses. Two of the houses (Washburne and Callahan) had high-efficiency furnaces, and the furnaces in the Grant home were recent-vintage medium-efficiency systems.

However, blower door test results for the three houses indicated high rates of air movement into the structures. This can be particularly problematic in historic homes, because high air infiltration is closely linked not only to higher energy bills and reduced comfort, but

also to increased moisture movement into walls and attics and higher indoor humidity levels during the summer. Both types of moisture movement can reduce the longevity of a structure. Higher air infiltration can also increase the variation in relative humidity, potentially damaging the artifacts displayed within these historic homes.

While the blower door was operating, we inspected the houses to identify specific sites of air infiltration. We found that a number of the larger air leakage sites for the Washburne and Callahan houses were from the interior into the attic space. For the Grant and Washburne houses, much of the air infiltration came from the cellars. Identifying these locations was particularly useful because remediation work could be done fairly easily, since the attics and cellars are not visited by the public and are not historically sensitive features of the houses. Moreover, we found that attic insulation levels were below those recommended for residential buildings located in northwestern Illinois. We thought it best to address air infiltration deficiencies before installing additional insulation.

We also examined the integrity of the forced-air duct systems in the houses. We used a variation on the standard blower door test, called the subtraction method, which is per-

formed by making two blower door estimates. The first test is done with all duct vents open, as is normal. The second test is done after covering the vents with plastic to isolate the duct system. We subtracted the second blower door estimate from the first and adjusted for internal air entering the ducts using the table provided in the Minneapolis Blower Door Manual (see pages 43-44 of the May 1993 edition). This provided an estimate of how much air escapes from the ducts to unconditioned areas. All three houses were found to have leaky duct systems. The Washburne and Callahan houses presented greater problems, because the ducts traveled through the attic.

Energy Improvements and Initial Results

We used the HERS software to simulate and examine various improvement strategies and develop estimated cost savings. Discussions among the analysts and the IHPA staff focused on three proposed improvements:

1. Reduce overall air infiltration, with special emphasis on reducing air movement to attic areas in the Washburne and Callahan houses and movement to the cellars in the Grant and Washburne houses.

2. Tighten the duct systems, particularly where ducts travel through the attics in the Washburne and Callahan houses.

3. Install additional insulation in the attics of the houses after air sealing and duct tightening work has been done.

These energy efficiency measures were very unlikely to have any impact on the historic features of the houses.

In Illinois, the Illinois Home Weatherization Assistance program has developed a standard set of goals for reducing air infiltration. The goal for air sealing is to reduce infiltration by 20% if the initial CFM₅₀ is between 1,560 and 2,750, as it was in the Callahan house. (Other states set goals differently. For example, some states base their goals for air sealing on

house size. In Illinois, the goals are to move toward a building tightness limit without overdoing the cost.) The goal for air sealing is a 40% reduction if the initial CFM₅₀ is between 4,250 and 5,500, as it was in the Grant home. And the goal for air sealing is a 45% reduction in air leaks if the initial CFM₅₀ is 5,500 to 7,500, as was the case in the Washburne home. These goals were used in the simulations. The simulations also assumed the reduction of duct leakage to 100 CFM₂₅ and an increase in attic insulation to R-42 (see Table 3). The overall estimated savings for space heating if a typical family occupied these homes during a year, with typical weather and paying current fuel rates—this price is from before the rise in this past winter's energy bills—would be \$870 per year, or approximately 26%.

These estimated potential savings may seem large. However, some work was completed following our energy ratings. Weatherization and insulation work was done on the Grant home during the early fall of 2003. (Duct sealing was not done, in part because there were no ducts in the attic of the Grant home, but

also because some HVAC folks argued that the leakage to the basement space would dry it out and keep mold from developing. We thought this was an awfully expensive way to address that potential problem.) The analysts returned to the Grant home in November 2004 and performed an additional blower

Table 3. Air Infiltration Goals and Simulated Energy Use After Proposed Improvements

	Blower Door Results (CFM ₅₀)	Estimated Space Heating Usage (therms/year)	Estimated Space Heating Cost (\$/year)	HERS Score
Grant Home	2,740	2,390	902	74
Washburne House	3,990	3,050	1,100	74
Callahan House	1,900	900	365	80

Note: Table 3 is from the REM/Rate output assuming that all the upgrades we recommended were implemented.

SCREENING BEFORE RATING

Most property owners pause when confronted with the question of whether to hire an energy analyst. Everyone knows what to do when a furnace fails (call in an HVAC contractor) or a roof leaks (hire a roofing firm). But the standard response to the question of whether to bring in an energy analyst is to put off the decision and think about gathering more information.

One approach to gathering more information is to conduct a screening of past energy bills. This can be done using a packaged software tool, such as PRISM. (See www.princeton.edu/~marean/ for more information about PRISM.) However, many people may not wish to purchase software to screen only one historic home. An alternative is to separate energy consumption roughly into baseload and weather-sensitive usage and compare each usage to a benchmark. Here is a simple way to do it for natural gas-heated homes.

Baseload natural gas usage can be computed by taking the lowest

monthly usage in therms of natural gas for a typical month in which space heating is not used and dividing by the number of billing days in the month. June, July, and August are usually good months to choose. Atypical months would be those with unusual schedules, such as holiday closings, or unusual activities, such as renovation work. Baseload usage is expressed as therms per day. If usage is greater than 1 therm per day, you should investigate hot water usage or baseload equipment and appliances.

Space-heating usage can be estimated by taking a typical midwinter month's natural gas consumption in therms, subtracting out the daily baseload energy usage multiplied by the number of billing days in that month, and dividing by the product of the number of heating degree-days in that month (to base 65°F) and the floor area of conditioned space expressed in ft². Multiply the result by 100,000 to

convert from therms to Btu. The final estimate will be given in units of Btu/HDD/ft². A result of more than 10 generally indicates that it will pay to conduct an energy audit or rating. (The Callahan house would not have been rated if we followed this method.)

Most natural gas utilities include the therms of gas consumed on the monthly energy bill, along with the number of billing days in the month. Often heating degree-days are also included on the bill. If heating degree-days are not listed on the bill, they can usually be obtained for a nearby major city from the National Climatic Data Center (www.ncdc.noaa.gov). The floor area of conditioned space should not include unconditioned areas, such as side porches or basements. If the utility estimates natural gas usage for some months, one should sum the usage and billing days for two or more months so that the calculation is based on usage going from one actual meter reading to another actual meter reading.



The Callahan house was built in 1891. It is now used as offices for the Illinois Historic Preservation Agency in Galena.

door test. Natural gas bills for the winter of 2003–04 were also analyzed. The blower door test showed that the weatherization work reduced air infiltration by 8%—from 4,565 CFM₅₀ to 4,198 CFM₅₀. The analysis of the natural gas bills showed that space-heat energy consumption was reduced by 12%—from 16 to 14 Btu/HDD/ft². This translates to 470 therms of natural gas saved at the Grant home during a winter with typical weather (7,376 heating degree-days).

Applying Rating Methods Elsewhere

Applying the tools and methods of home energy ratings to historic homes enables state agencies or conservators to identify and capture energy savings while still maintaining—if not improving—the comfort of the home. The elements of the approach that can be used in the context of historic homes are as follows:

- Conduct a simple screening of the utility bills for space-heating fuel use to roughly evaluate the potential for energy efficiency improvements. If space-heating energy consumption is high relative to similar homes—or if it is greater than 10 Btu/HDD/ft²—call in a home energy

rater or a whole-house energy auditor.

- When conducting an energy analysis, be sure to include a blower door test. Identify and list the locations of air infiltration for possible action. Pay particular attention to air leakage to the attic or from crawlspaces and cellars. If furnace and A/C ducts travel through unconditioned spaces, such as the attic, measure how tight those ducts are, using a Duct Blaster or applying the blower door subtraction method discussed above. If possible, conduct an infrared scan of the home while the blower door is operating to identify air movement within the walls and ceilings.

- Capture the improvement opportunities in areas that do not impinge on historic features. These will probably include sealing air bypasses to the attic, crawlspace, and cellar; tightening all furnace and A/C ducts that travel through unconditioned spaces, such as the attic and crawlspace; weatherstripping doors and hatches to unconditioned spaces; and adding insulation to the attic. Blowing insulation into the walls usually requires opening portions of the walls either from the inside or the outside. In either case, the goal is to do what can be done without touching the

parts of historic homes that are seen by the public and without compromising the historic elements of the home.

- Evaluate how best to address opportunities associated with the historic features of the home. If, for instance, the windows leak badly, this should be addressed the next time the windows are reconditioned or refinished. Certainly the building systems that present the historic character to the public must be preserved. However, the energy analysis may identify many opportunities that result from the house's age, and that should be included in a restoration program in any case. Evaluating how best to address these opportunities is just good stewardship.

Certainly, reducing energy waste in one, two, or three homes will not generate a sizable resource dividend to pass on to future generations. However, the actions that can be taken within the restrictions of a historic home in which preservation is an absolute priority can be implemented in existing homes with fewer constraints. Moreover, since thousands of visitors pass through a historic home like Grant's each year (between 65,000 and 85,000 people visit the Grant home each year, according to Dan Trindle of the IHPA Galena Historic Site), these homes provide opportunities to educate the public about energy efficiency and sustainability. Assisting in the preservation of historic homes is also a great application of the tools and talents of home energy raters.



*James Cavallo is principal at Kouba-Cavallo Associates, Incorporated, and an associate editor at **Home Energy**. Chet Marszalek is a home energy rater in Galena, Illinois, and owner of Galena People Heaters.*

FOR MORE INFORMATION:

Information on the Grant and Washburne homes is available at www.GrantHome.com.